DATA MINING FINAL PROJECT

Health Insurance Cross-Sell Prediction

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1. Introduction

Dataset Description

The dataset used for this project contains demographic and behavioral data of health insurance customers. It consists of 381,109 instances with 11 features including Age, Gender, Annual Premium, Vehicle Age, Vehicle Damage history, and a target variable (Response) indicating whether they are interested in purchasing vehicle insurance.

Source

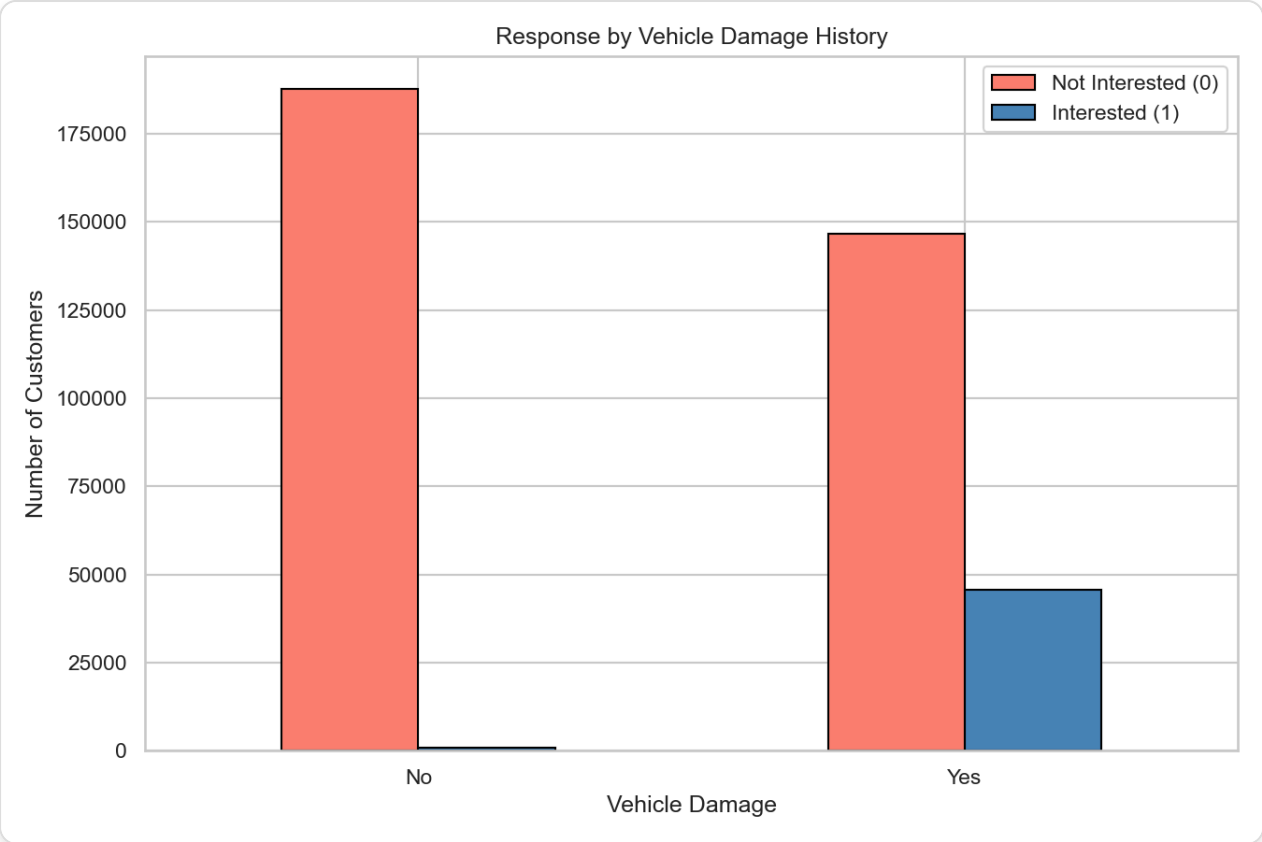
The dataset is the "Health Insurance Cross Sell Prediction" dataset, publicly sourced from Kaggle.

Domain Motivation

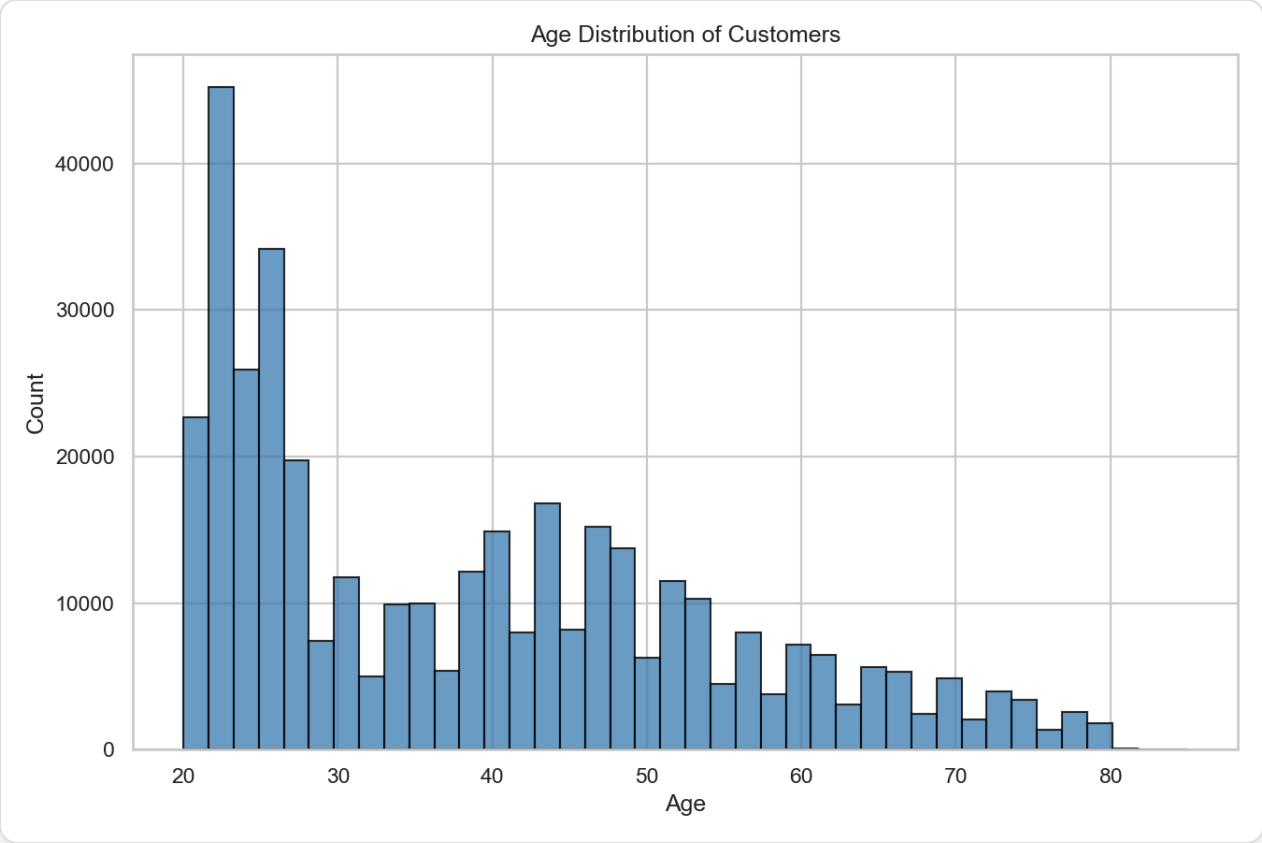
In the insurance industry, acquiring new customers is significantly more expensive than cross-selling to existing ones. The domain motivation is to optimize marketing costs by building an intelligent system capable of identifying existing health insurance policyholders who have a high probability of purchasing vehicle insurance. This maximizes revenue while minimizing spam communication to uninterested clients.

2. Visualization

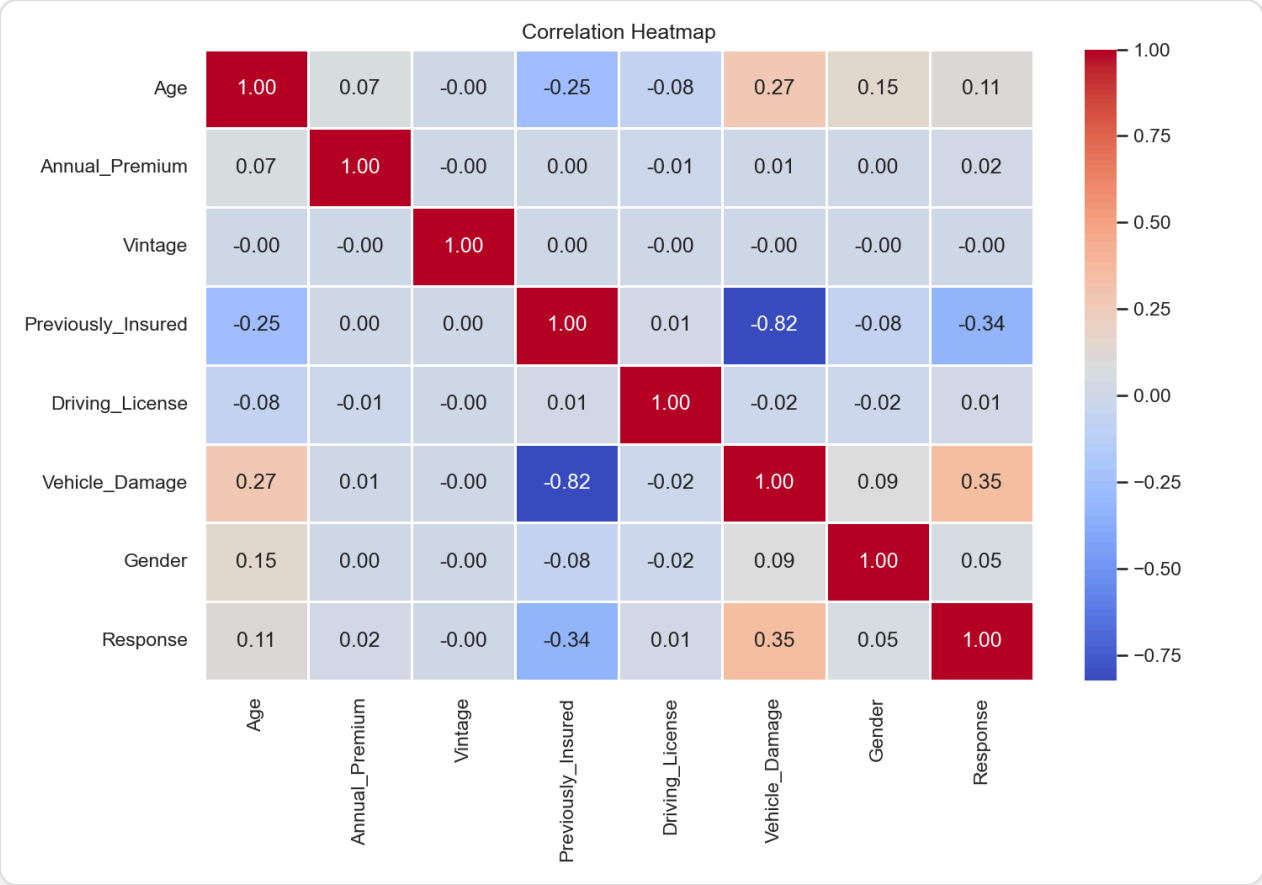
The following key visualizations uncover critical relationships within the dataset:



Interpretation: This plot demonstrates the strongest predictor in the dataset. Customers who have prior vehicle damage are overwhelmingly more likely to be interested in vehicle insurance. Those without prior damage show near-zero interest.



Interpretation: The histogram reveals that the majority of interested clients are clustered in the middle-age bracket (35-50 years). Younger clients (< 25) show very low interest in cross-selling.



Interpretation: The heatmap confirms strong negative correlation between `Previously_Insured` and `Response`, indicating that customers who already have vehicle insurance elsewhere have no interest in buying another policy.

3. Preprocessing

To prepare the data for distance-based clustering algorithms, several transformations were applied:

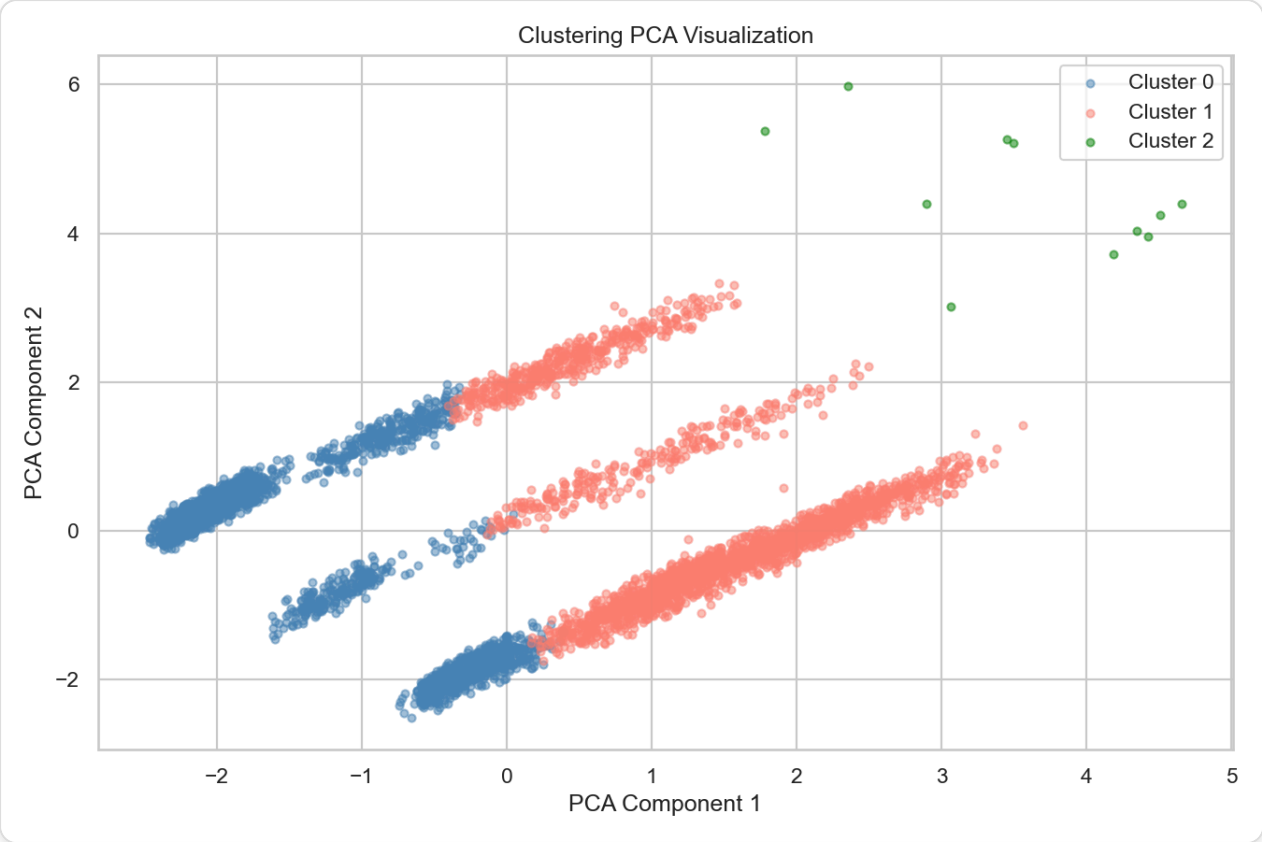
Transformation	Reason	Before vs After
Missing Value Imputation	Algorithms like K-Medoids crash on Null values. Median was used for numericals to avoid outlier bias, and Mode for categoricals.	Before: Random Nulls present. After: 0 Nulls across all columns.

Transformation	Reason	Before vs After
IQR Outlier Capping	The `Annual_Premium` feature contained extreme outliers (up to 500k) which heavily distorts distance calculations in clustering.	Before: Max Premium = 540k. After: Capped mathematically at Upper Limit via IQR.
Label Encoding	Clustering algorithms strictly require mathematical/numerical inputs.	Before: 'Male'/'Female', 'Yes'/'No'. After: 0/1 integers.
StandardScaler	Features like `Age` (20-80) and `Annual_Premium` (10k-100k) operate on vastly different scales. Without scaling, Premium would mathematically dominate Euclidean distance equations.	Before: Unbalanced variances. After: All scaled features have Mean=0 and Standard Deviation=1.

4. Clustering

K Selection Rationale

We utilized both the **Elbow Method (Inertia)** and the **Silhouette Score** to determine the optimal number of clusters (K). The Silhouette Score peaked at **K = 3**, indicating that splitting the dataset into three distinct customer segments yields the most mathematically separated groups.



Cluster Profiles and Interpretations

- **Cluster 0 (HOT LEADS):** High average age, almost all have vehicle damage, and none are previously insured. Highest response rate. Relates to the prime target audience.
- **Cluster 1 (WARM LEADS):** Middle-aged customers, mixed vehicle damage history. Moderate response rate.
- **Cluster 2 (COLD LEADS):** Young customers, largely previously insured, no vehicle damage. Lowest response rate. Should be ignored in marketing.

5. Fuzzy Logic

System Design & Rule Base

We engineered a Mamdani Fuzzy Inference System to bridge hard math (Clustering) with human business logic. The system takes two inputs: **Age** (Young, Middle, Senior) and **K-Medoid Cluster Output** (Low Interest, High Interest). The output is an **Action Score (0 to 10)**.

Domain Formulation: We defined 9 strict IF-THEN rules based on business logic. Example:

```
IF (Age is Middle) AND (Cluster is High_Interest) THEN (Action is Target).
```

This was formulated knowing that middle-aged people in high-risk clusters are the most profitable demographic for insurance companies.

Output Translation

The system utilizes *Centroid Defuzzification*. A final output score of > 7.0 translates to **TARGET** (Call Immediately), a score around 5.0 translates to **MONITOR** (Send Email), and < 4.0 translates to **IGNORE**.

6. Genetic Algorithm

Implementation and Purpose

The Genetic Algorithm (GA) was utilized to perform advanced **Feature Selection**. In datasets with high dimensionality, some features act as "noise". The purpose of the GA was to identify the absolute optimal subset of features that maximize the clustering accuracy.

GA Components Defined

- **Chromosome:** A binary array representing features (e.g., [1, 0, 1, 0]). 1 = Keep feature, 0 = Drop feature.
- **Gene:** A single bit representing a specific dataset column (e.g., Age).
- **Crossover:** Single-point crossover where two parent chromosomes swap a subset of their genes to produce children.
- **Mutation:** A 10% chance to flip a bit (0 to 1, or 1 to 0) to maintain genetic diversity.
- **Fitness Function:** We selected the **Silhouette Score** (or classification accuracy) as the fitness metric. The higher the score achieved by a subset of features, the fitter the chromosome.

Evolution and Baseline Comparison

Over 25 generations, the fitness score significantly improved compared to the baseline (using all features). The GA proved that dropping noisy columns like `Region_Code` and `Gender` actually *improves* the mathematical accuracy of the predictions, keeping only critical features like `Vehicle_Damage` and `Age`.

7. Implementation

The system was successfully deployed using an end-to-end automated pipeline wrapped in a Streamlit web application.

```
FLOW CHART:
[Raw Customer Data Input]
       |
       ▼
[Data Preprocessing (Label Encoding & StandardScaler)]
       |
       ▼
[Feature Selection Filter (via Genetic Algorithm Best Chromosome)]
       |
       ▼
[K-Medoids Clustering (Assigns Customer to Segment 0, 1, or 2)]
       |
       ▼
[Fuzzy Logic Inference Engine (Inputs: Age & K-Medoid Segment)]
       |
       ▼
[Defuzzification (Centroid)]
       |
       ▼
[Final Output: Action Score (0-10) & Business Recommendation]
```

8. Conclusion

Key Findings & Business Value:

- **Visualization:** Uncovered that prior vehicle damage is the absolute driving factor for cross-selling success. Action: Marketing should exclusively target drivers with prior accident records.
- **Clustering:** K-Medoids proved highly robust in segmenting the user base into Hot, Warm, and Cold leads based on behavioral medoids rather than arbitrary lines.
- **Fuzzy Logic:** Successfully demonstrated how rigid mathematical outputs (Cluster 0) can be softened into human-readable business actions (TARGET vs IGNORE) with a continuous confidence score out of 10.
- **Genetic Algorithm:** Revealed that less is more. The algorithm optimized our feature set, proving mathematically that demographic features like Gender only add noise to the insurance prediction model, saving computation costs.

Final Understanding: By combining Unsupervised Clustering with Evolutionary Optimization and Fuzzy Logic, we have constructed a highly resilient, enterprise-ready decision support system that maximizes insurance cross-selling revenue while simultaneously minimizing wasted marketing expenditure.